

Gasol Shift Tracker

From dashboard insight to data capture

An Excel and VBA system for shift-level data at a petrol filling station

Data Story | Prepared by [DICKSON] Gasol Petroleum case study

1. Executive summary

Gasol Petroleum is a petrol filling station selling two products, premium petrol (PMS) and diesel (AGO), across two shift teams. I first built a Power BI dashboard to understand operations. It showed where the business was healthy and where the records were weak. Missing dates, stock that did not reconcile and small banking differences all pointed to one root cause: data was being captured inconsistently.

To fix this at the source I built **Gasol Shift Tracker**, a macro-enabled Excel workbook with three guided entry forms, three structured databases, dropdown-controlled inputs and automatic calculations for stock variance, banking variance and revenue. Every shift now produces consistent, analysis-ready records.

GHS 5.87M Total revenue	441.08K L Litres sold	GHS 179.03K Total expenses	3.05% Expense-to-sales ratio
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2. Business context and problem

A fuel station runs on three flows that must agree with each other: **fuel** (what was delivered, dipped and sold), **cash** (what was collected, banked or given on credit) and **costs** (payroll, maintenance, head-office charges and more). When any one of them is recorded loosely, the others cannot be checked against it. The business questions were practical ones:

- Is stock in the tanks matching what the records say it should be?
- Does the cash banked match the cash expected from sales?
- Which shifts, attendants and products drive revenue?
- Where do costs go, and how much credit is still owed?

3. What the dashboard revealed

The Power BI report has four pages: an overview, sales and operations, inventory and variance, and cash flow, banking and credit. It uses slicers for month, shift-group and product.

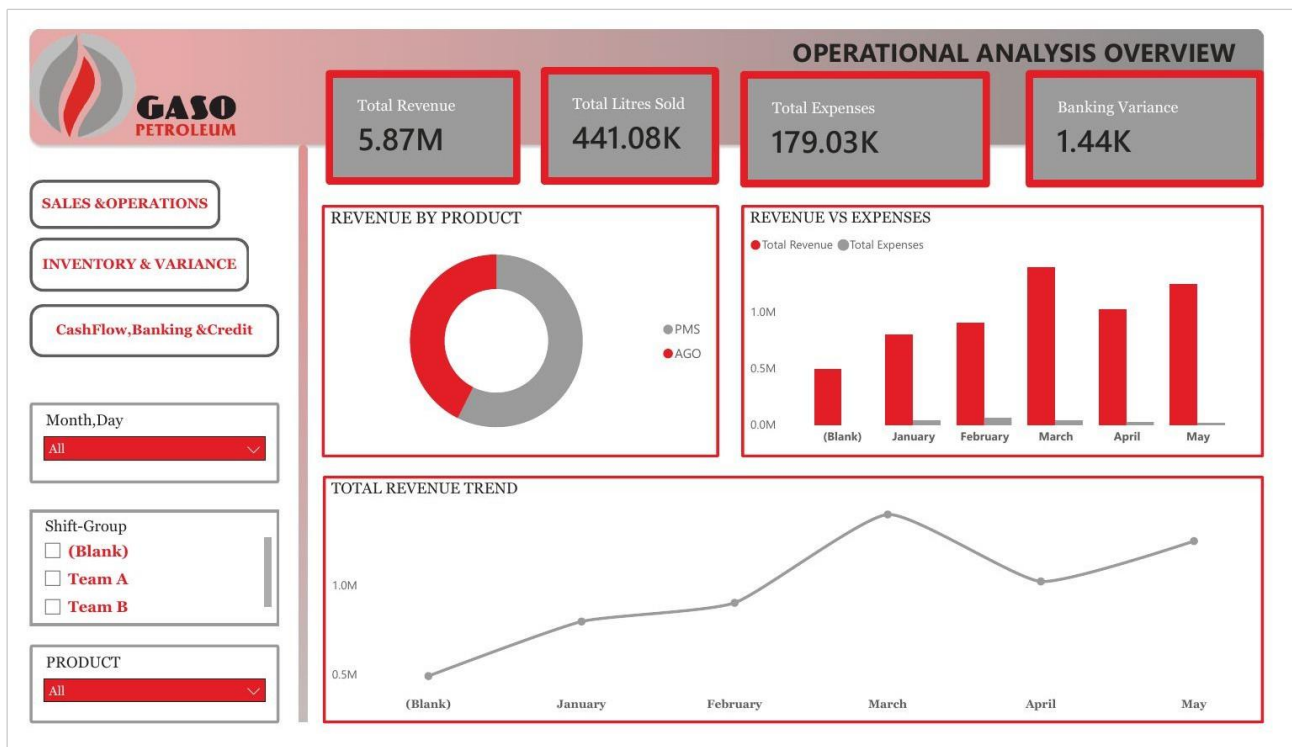


Figure 1. Operational Analysis Overview: headline KPIs, revenue by product, revenue vs expenses and revenue trend.

Sales and performance

- Revenue of GHS 5.87M came from 441.08K litres at an average selling price of GHS 13.30 per litre.
- PMS contributes roughly 60% of revenue and AGO roughly 40%, judging from the product split.
- Revenue peaked in March at about GHS 1.39M, dipped in April, then recovered in May.
- Team B and Team A are close in revenue, each near GHS 3M. Abigail Bawah leads the attendant ranking.

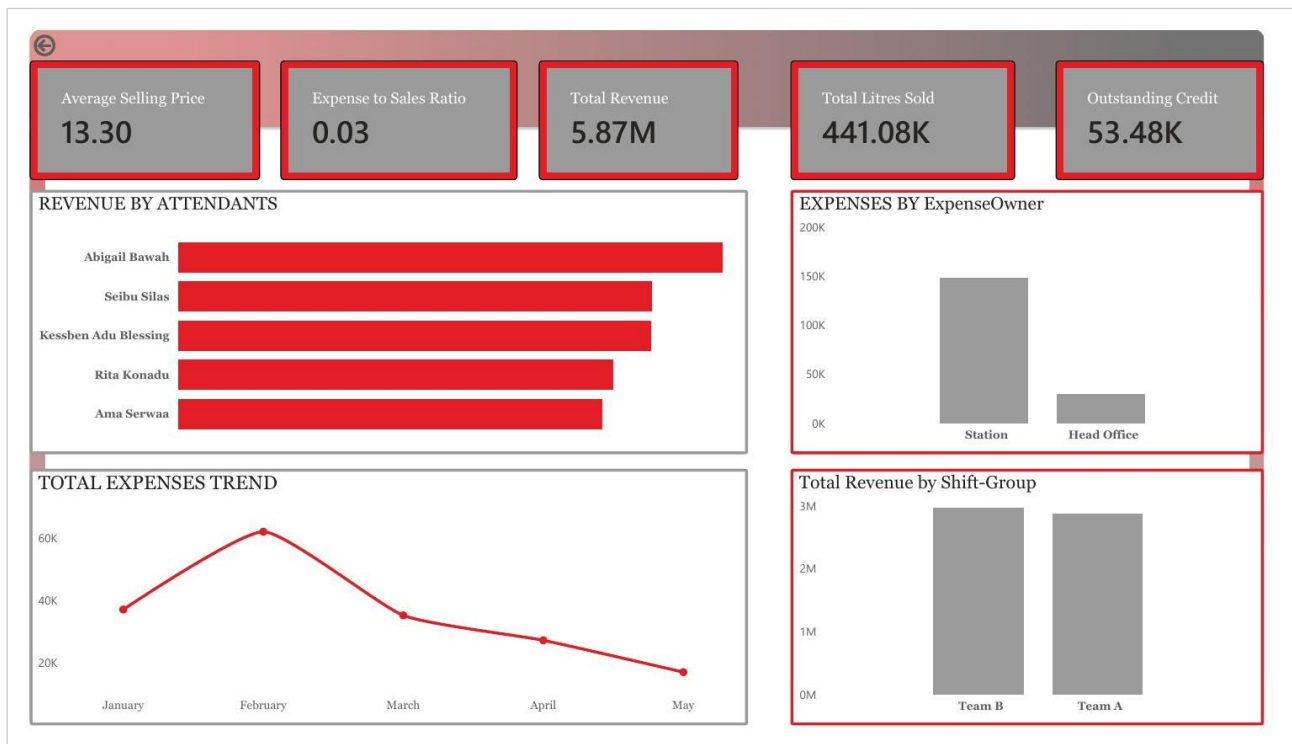


Figure 2. Sales and Operations: revenue by attendant, expenses by owner, expense trend and revenue by shift-group.

Cost control

Expenses are low relative to sales. Across the five months with dated records, the ratio of expenses to cash sales fell from about 7.0% in February to 1.4% in May.

Month	Cash sales (GHS)	Expenses (GHS)	Expense ratio
January	784,735	37,175	4.7%
February	894,191	62,166	7.0%
March	1,385,485	35,307	2.5%
April	1,019,814	27,300	2.7%
May	1,232,860	17,083	1.4%

Payroll is the largest category at roughly 59% of all spend, and station-owned costs make up about 84% of expenses against 16% for head office. Ratios are derived from the dashboard's monthly table.

Inventory and tank variance

Deliveries totalled 448,790 litres. Net tank variance was -3,222 litres, or about 0.72% of deliveries. PMS accounts for almost all of the loss. Tank variance is a key early warning for leakage, metering faults or theft, so even a small net figure deserves follow-up.

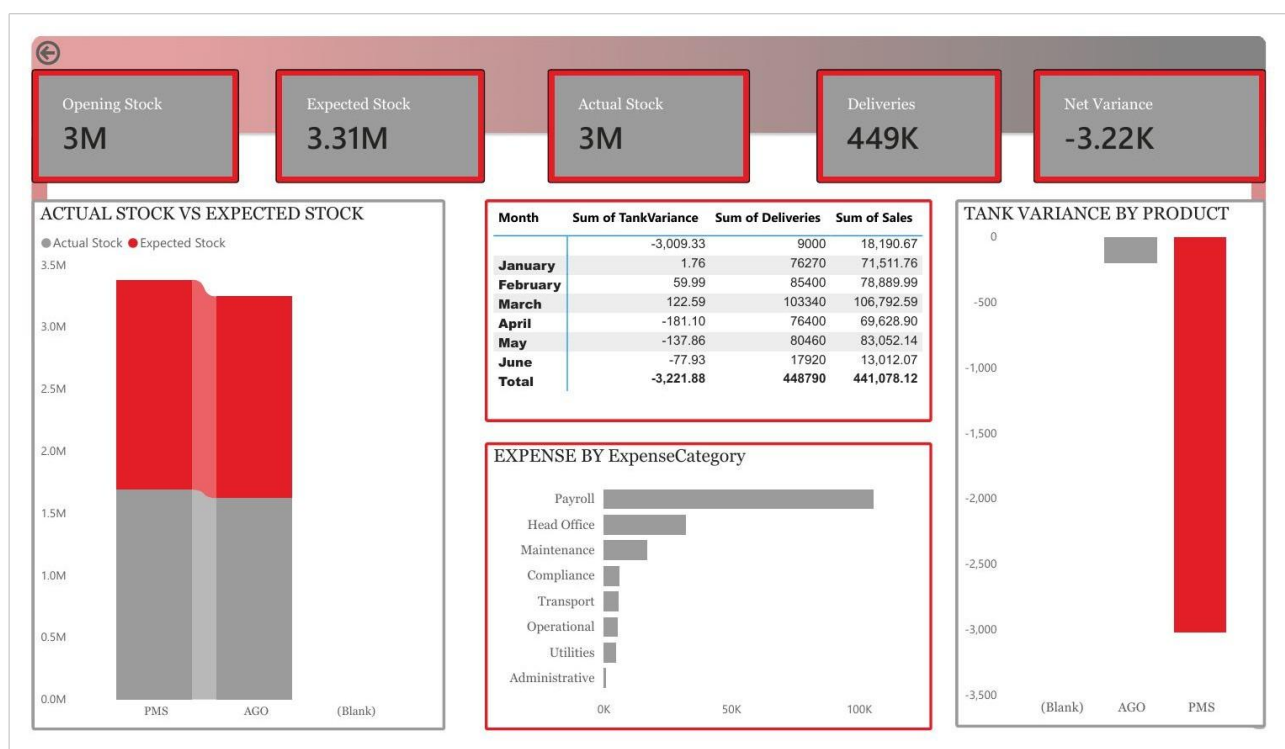


Figure 3. Inventory and Variance: opening, expected and actual stock, monthly variance table and variance by product.

Cash, banking and credit

Cash sales were GHS 5.32M and credit sales GHS 53.48K, about 1% of the total. Banking variance was small at GHS 1.44K (about 0.03% of cash sales) but peaked in April. Outstanding credit is concentrated: Carbon Logistics owes the most, at roughly GHS 31K of the total GHS 53.48K.

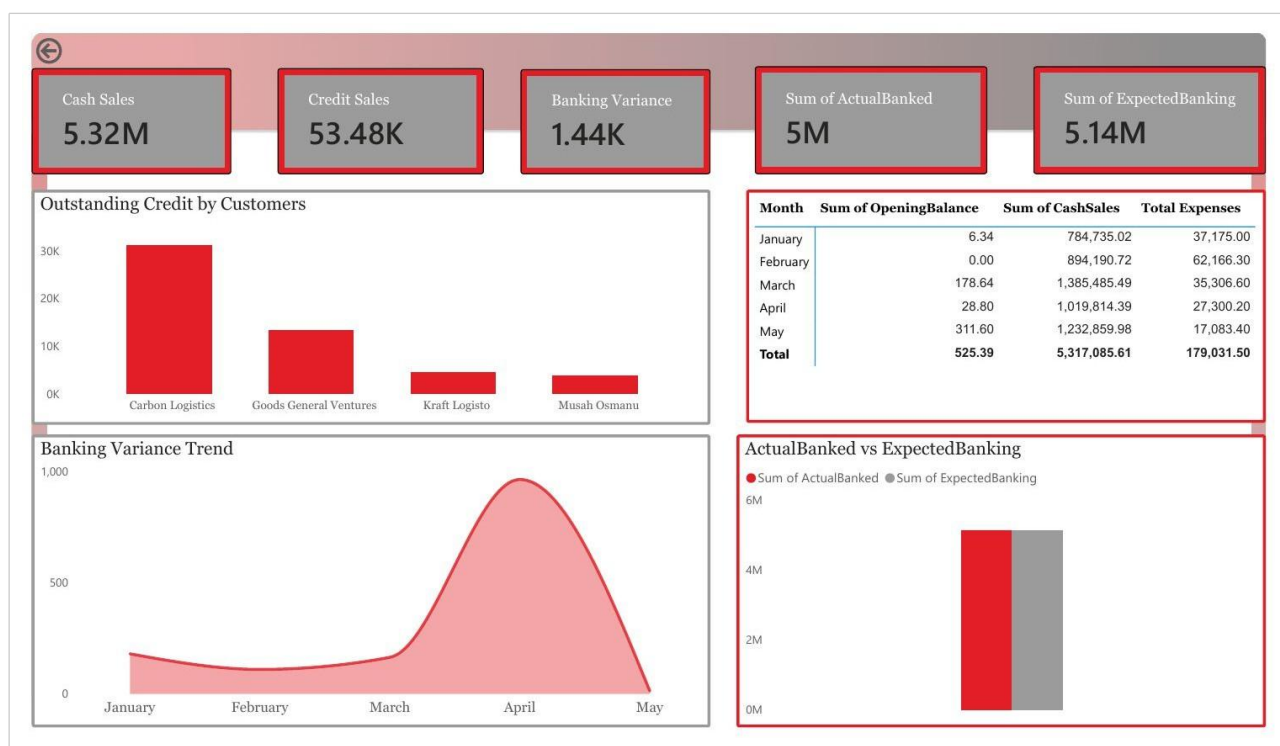


Figure 4. Cash Flow, Banking and Credit: outstanding credit by customer, banking variance trend and actual vs expected banking.

4. The gap behind the numbers

Revenue shows GHS 5.87M, but cash plus credit sales add up to only about GHS 5.37M. The difference of roughly GHS 0.5M closely matches the "(Blank)" period in the revenue charts, which suggests records with no valid date.

The inventory table has a similar blank row carrying 9,000 litres delivered and a -3,009 litre variance, which drives most of the net loss.

This was the turning point. The dashboard was only as reliable as the data feeding it. Blank dates, free-typed names and uncalculated variances were weakening every chart. A better dashboard would not solve that, but a better capture process would.

5. The solution: Gaso Shift Tracker

I designed the tracker so that a shift supervisor can record a day's activity in minutes, without touching formulas or database sheets.

Module	Entry sheet	Database table	Calculated automatically
Fuel sales	Entry_FuelSales	DB_FuelSales (14 fields)	Expected stock = opening + deliveries - litres sold Stock variance = actual - expected Revenue = litres x unit price
Cash and banking	Entry_CashBanking	DB_CashBanking (11 fields)	Banking variance = actual banked - expected banking
Expenses	Entry_Expenses	DB_Expenses (8 fields)	Category and owner captured through controlled lists

A **Lookups** sheet feeds the dropdowns for shift group, attendant, product, expense category, expense owner and credit customer. A **Dashboard** sheet shows live totals for revenue, litres, expenses, banking variance and stock variance using structured-table formulas, so it updates as soon as an entry is saved.

The entry screens

NEW FUEL SALES ENTRY

Date	25/07/2026	e.g. 23/07/2026
Shift-Group	Team B	
Attendant	Rita Konadu	
Product	AGO	
Opening Stock (L)	24000	
Deliveries (L)	14000	
Actual Stock (L)	25000	measured/dipped at end of shift
Litres Sold	13000	
Unit Price (GHS)	14	
Expected Stock (L)	25000	
Stock Variance (L)	0	
Revenue (GHS)	182000	

SAVE ENTRY

Figure 5. Fuel sales entry, with live expected stock, stock variance and revenue.

In Figure 5, a test AGO entry shows the logic at work: 24,000 L opening + 14,000 L delivered - 13,000 L sold gives 25,000 L expected. The dipped stock matches, so variance is 0, and 13,000 L at GHS 14 gives revenue of GHS 182,000.

NEW CASH & BANKING ENTRY

Date		
Shift-Group		
Opening Balance (GHS)	0	
Cash Sales (GHS)	0	
Credit Sales (GHS)	0	leave blank if no credit sale
Credit Customer		from bill reconciliation
Expected Banking (GHS)	0	from bank teller slip
Actual Banked (GHS)	0	
Banking Variance (GHS)	0	

SAVE ENTRY

Figure 6. Cash and banking entry, with a banking variance field and a credit-customer prompt.

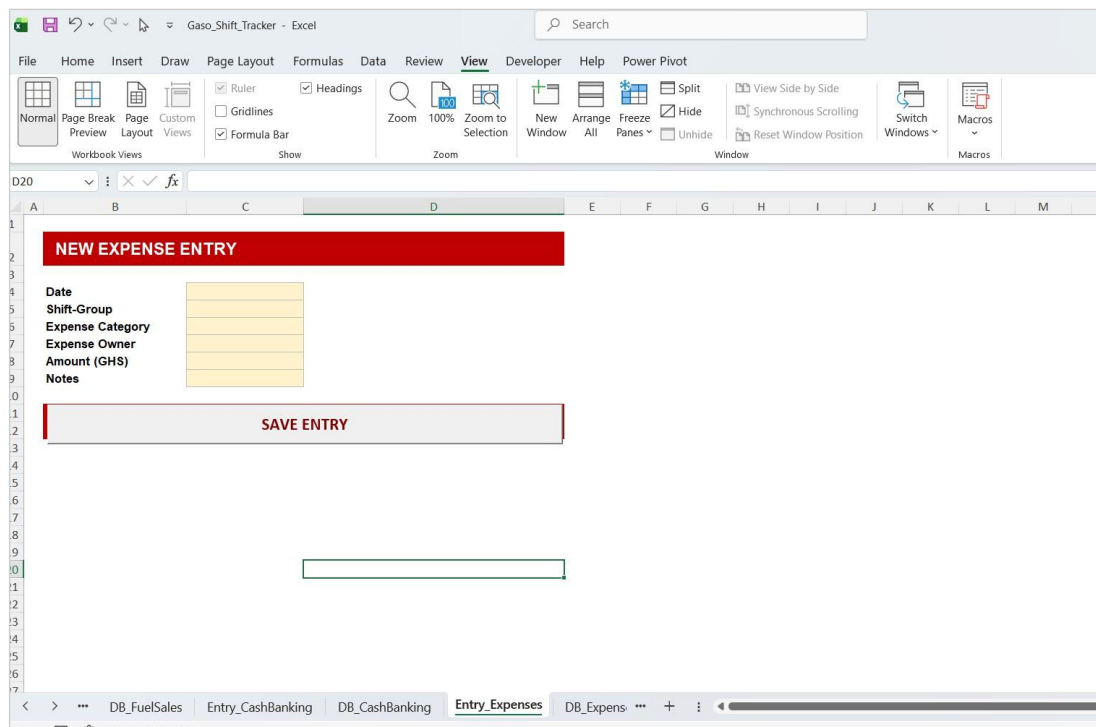


Figure 7. Expense entry, with category, owner, amount and notes.

6. How the macros work

Three VBA procedures, `SaveFuelSales`, `SaveCashBanking` and `SaveExpense`, sit behind the SAVE ENTRY buttons. Each one follows the same pattern:

1. **Validate:** refuse to save if required fields are blank (date and shift for every form, plus attendant and product for fuel, or category and amount for expenses), with a clear message.
2. **Locate:** `NextEmptyRow` finds the first free row in the database sheet.
3. **Identify:** `NextID` assigns the next sequential ID by reading the current maximum.
4. **Write:** copy each form field to the matching column, apply a consistent DD/MM/YYYY date format and stamp an `EnteredOn` timestamp for audit.
5. **Extend:** resize the Excel Table so formulas, charts and Power BI pick up the new row.
6. **Reset:** clear the input cells ready for the next entry, keeping the date on the fuel form, and confirm the saved Shift ID.

7. Business value

- **Cleaner data:** mandatory fields and dropdowns remove blank dates and spelling variations at the point of entry.
- **Earlier detection:** stock and banking variances are computed the moment a shift is recorded, not discovered months later.
- **Accountability:** every record carries a Shift ID, shift group, attendant and entry timestamp.
- **Faster reporting:** the three tables are structured for direct connection to Power BI, so the dashboard refreshes from reliable data.
- **Low cost:** it runs in Excel, which the station staff already know, with no new software or licences.

8. Limitations and next steps

- The workbook is single-user and has no edit or delete screen. Corrections are made directly in the database tables.
- Variances are recorded but not yet flagged. A conditional alert for any variance beyond a set threshold would turn the tracker into an active control.
- Cash, fuel and expense entries are linked by date and shift only. A shared shift key would allow automatic cross-checks of revenue against cash plus credit.
- Next steps: add the Power BI refresh from these tables, protect the database sheets, and migrate to a shared database if several stations are added.

9. Skills demonstrated

Business and operations analysis | Data modelling and structured tables | Excel VBA automation | Data validation and quality control | Power BI dashboard design | Data storytelling and documentation.